

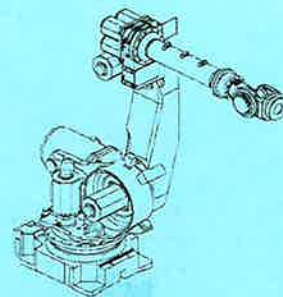


OAKLAND COMMUNITY COLLEGE

ROBOTICS/AUTOMATED SYSTEMS TECHNOLOGY

ROB 2140

LAB SIGN-OFF SHEET



Name: Tom Desrosiers

FALL 2025

Computer: T4-4

DO NOT LOSE THIS SHEET, NO DUPLICATE RECORD KEPT.

**ALL LABS ARE DUE BY THE END OF SESSION 14, DECEMBER 2.
RETURN SIGN-OFF SHEET BEFORE OR ON SESSION 14.**

**LABS MUST BE TESTED PRIOR TO SUBMITTING.
LABS ARE PASS/FAIL. ONE ERROR IS EQUIVALENT TO A
FAIL. LABS CANNOT BE RESUBMITTED.**

LAB POINTS EQUATES TO PERCENTAGE OF THE COURSE GRADE.

ASSIGNMENT	POINTS	INITIALS	DATE
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Studio 5000 Ladder Logic Labs:

MOVE INSTRUCTION, PART 1	0.5	JPS	9-2
MOVE INSTRUCTION, PART 2 <i>Signed Off with HMI</i>	1	JPS	9-20
TIMED PROCESS OUTPUT <i>Signed Off with HMI</i>	1.5	JPS	10-7
TEMPERATURE CONVERSION with INDICATORS <i>Signed Off with HMI</i>	1.5	JPS	10-7
☐ BORE	3		
☐ TORQUE 7	1.5		
☐ ANALOG INPUT <i>Signed Off with HMI</i>	1.5		

Instructors Use

Completed Lab _____

Final Grade _____

ASSIGNMENT	POINTS	INITIALS	DATE
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Siemens Ladder Logic Labs:

MOVE INSTRUCTION, PART 1	0.5	JPS	9-9
MOVE INSTRUCTION, PART 2	1	JPS	9-16
TIMED PROCESS OUTPUT	1.5	JPS	9-23
TEMPERATURE CONVERSION with INDICATORS	1.5	JPS	10-14
ANALOG INPUT	1.5		

Studio 5000 Function Block Labs:

SS1	1.5	JPS	10-14
WASH TIMER ON	1.5	JPS	10-14
TIMED PROCESS OUTPUT	1.5	JPS	10-21

Siemens Function Block Labs:

SS1	1.5	JPS	10-21
WASH TIMER ON	1.5	JPS	10-21
TIMED PROCESS OUTPUT	1.5	JPS	10-28

Studio 5000 Structured Text Labs:

SS1	1.5	JPS	10-28
WASH TIMER ON	1.5	JPS	10-28
TEMPERATURE CONVERSION with INDICATORS	1.5	JPS	10-28

Siemens Structured Text Labs:

SS1	1.5	JPS	10-28
WASH TIMER ON	1.5	JPS	10-28
TEMPERATURE CONVERSION with INDICATORS	1.5		

Studio 5000 Add On Instruction Labs:

ANALOG SCALING	1.5		
MIG WELD CONTROLLER	1.5		

7-11-2015

ASSIGNMENT	POINTS	INITIALS	DATE
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PLC and FANUC Robot

CREATE NEW ROBOT ETHERNET/IP CONNECTIONS	1.5		
CREATE NEW MACHINE ETHERNET/IP CONNECTIONS			
IMPORT TAGS	1		
ADD REQUIRED USER OPERATOR PANEL SIGNALS			
SAVE AS AN EXPORT FILE	1		
IMPORT A PLC FILE			
PROGRAMMING START AND STOP FOR ROBOT CONTROL	1.5		
PROGRAMMING FAULT MONITORING AND RESET			
PROGRAMMING STATUS FOR READY TO RUN\RUNNING			
PROGRAMMING FOR STYLE SELECT	2		
PROGRAMMING FOR STANDARD ROBOT I/O TO PLC I/O	2		
PROGRAMMING FOR ROBOT GROUP I/O <i>session 11</i>			

HMI LABS: CHECK THE PROPERTIES OF ALL HMI OBJECTS AND TEST FOR PROPER OPERATIONS AND APPEARANCE. ONE MISTAKE IS A FAILED HMI PROJECT.

FactoryTalk View ME HMI Labs:

MOVE INSTRUCTION	3	<i>JPS</i>	<i>9-27</i>
TIMED PROCESS OUTPUT	4	<i>JPS</i>	<i>10-7</i>
TEMPERATURE CONVERSION with INDICATORS	4	<i>JPS</i>	<i>10-7</i>
ANALOG INPUT	4		

ROB 2140 Session Assignments**Fall 2025**

Studio 5000 Ladder Logic Labs

MOVE INSTRUCTION, PART1	Session 1
MOVE INSTRUCTION, PART 2	Session 2
TIMED PROCESS OUTPUT	Session 2
TEMPERATURE CONVERSION with INDICATORS	Session 4
BORE	Session 8
TORQUE	Session 10
ANALOG INPUT	Session 11

Siemens Ladder Logic Labs

MOVE INSTRUCTION, PART 1	Session 1
MOVE INSTRUCTION, PART 2	Session 2
TIMED PROCESS OUTPUT	Session 2
TEMPERATURE CONVERSION with INDICATORS	Session 4
ANALOG INPUT	Session 11

Studio 5000 Function Block Labs

SS1	Session 3
WASH TIMER ON	Session 3
TIMED PROCESS OUTPUT	Session 4

Siemens Function Block Labs

SS1	Session 3
WASH TIMER ON	Session 3
TIMED PROCESS OUTPUT	Session 4

Studio 5000 Structured Text Labs

SS1	Session 5
WASH TIMER ON	Session 5
TEMPERATURE CONVERSION with INDICATORS	Session 6

Siemens Structured Text Labs

SS1	Session 5
WASH TIMER ON	Session 5
TEMPERATURE CONVERSION with INDICATORS	Session 6

Studio 5000 Add On Instruction Labs

ANALOG SCALING	Session 12
MIG WELD CONTROLLER	Session 12

Studio 5000 and FANUC Robotics

CREATE NEW ROBOT ETHERNET/IP CONNECTIONS	Session 7
CREATE NEW MACHINE ETHERNET/IP CONNECTIONS	Session 7
IMPORT TAGS	Session 7
ADD REQUIRED USER OPERATOR PANEL SIGNALS	Session 7
SAVE AS AN EXPORT FILE	Session 8
IMPORT A PLC FILE	Session 8
PROGRAMMING START AND STOP FOR ROBOT CONTROL	Session 9
PROGRAMMING FAULT MONITORING AND RESET	Session 9
PROGRAMMING STATUS FOR READY TO RUN/RUNNING	Session 9
PROGRAMMING FOR STYLE SELECT	Session 10
PROGRAMMING FOR STANDARD ROBOT I/O TO PLC I/O	Session 11

FactoryTalk View ME HMI Labs

MOVE INSTRUCTION	Session 2
TIMED PROCESS OUTPUT	Session 3
TEMPERATURE CONVERSION with INDICATORS	Session 4
ANALOG INPUT	Session 11

Session Dates

Session 1	August 26, 2025
Session 2	September 2, 2025
Session 3	September 9, 2025
Session 4	September 16, 2025
Session 5	September 23, 2025
Session 6	September 30, 2025
Session 7	October 7, 2025
Session 8	October 14, 2025
Session 9	October 21, 2025
Session 10	October 28, 2025
Session 11	November 4, 2025
Session 12	November 11, 2025
Session 13	November 18, 2025

Thanksgiving Recess

Session 14	December 2, 2025	ALL LAB DUE
Session 15	December 9, 2025	FINAL